MTH 4300 Activity Day 1 **NAME:**
Intro to C++

Instructions: Complete this activity after reading through the "Intro to C++ and control structures preview" notes. Use **onlinegdb.com** to run and test your code.

This paper version is for your reference; you can complete the activity directly on Amplify at:
`https://student.amplify.com/greeting/6973cb8dfa63cde51d27aff7?checkAmplifyLogin=true`

Class code: `AJQQ9R` Invite link: `https://learning.amplify.com/join/#/AJQQ9R`

1. *The Buggy Average.* The notes include a program that calculates the average of two quiz scores:

```cpp
#include <iostream>
int main() {
    int score1 = 85;
    int score2 = 90;
    int average = (score1 + score2) / 2;
    std::cout << "Average: " << average;
    return 0;
}
```

   (a) What output do you *expect* this program to produce?

   (b) Run the program in OnlineGDB. What output do you *actually* get?

2. *Variables and Types.*

   (a) Suppose you have the following line of code:

   ```cpp
   int x = 10;
   ```

   Write a single line of code that updates the value of `x` to 12.

   (b) What value does `x` hold after the following line executes? Explain briefly.

   ```cpp
   int x = 10.7;
   ```

3. *Input and Output.* Write a complete C++ program that:

   - Asks the user for two numbers
   - Prints their **sum**

4. *Control Structures Preview.* The notes include three practice exercises for control structures (6a, 6b, 6c). Try **at least one** of them and write your code below.

   6a (while loop: print 10 down to 1) **6b** (for loop: print 10 down to 1) **6c** (even or odd)